Saturated free fatty acid sodium palmitate-induced lipoapoptosis by targeting glycogen synthase kinase-3β activation in human liver cells.

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BACKGROUND: Elevated serum saturated fatty acid levels and hepatocyte lipoapoptosis are features of nonalcoholic fatty liver disease (NAFLD). AIM: The purpose of this study was to investigate saturated fatty acid induction of lipoapoptosis in human liver cells and the underlying mechanisms. METHODS: Human liver L02 and HepG2 cells were treated with sodium palmitate, a saturated fatty acid, for up to 48 h with or without lithium chloride, a glycogen synthase kinase-3β (GSK-3β) inhibitor, or GSK-3β shRNA transfection. Transmission electron microscopy was used to detect morphological changes, flow cytometry was used to detect apoptosis, a colorimetric assay was used to detect caspase-3 activity, and western blot analysis was used to detect protein expression. RESULTS: The data showed that sodium palmitate was able to induce lipoapoptosis in L02 and HepG2 cells. Western blot analysis showed that sodium palmitate activated GSK-3β protein, which was indicated by dephosphorylation of GSK-3β at Ser-9. However, inhibition of GSK-3β activity with lithium chloride treatment or knockdown of GSK-3β expression with shRNA suppressed sodium palmitate-induced lipoapoptosis in L02 and HepG2 cells. On a molecular level, inhibition of GSK-3β expression or activity suppressed sodium palmitate-induced c-Jun-N-terminal kinase (JNK) phosphorylation and Bax upregulation, whereas GSK-3β inhibition did not affect endoplasmic reticulum stress-induced activation of unfolded protein response. CONCLUSIONS: The present data demonstrated that saturated fatty acid sodium palmitate-induced lipoapoptosis in human liver L02 and HepG2 cells was regulated by GSK-3β activation, which led to JNK activation and Bax upregulation. This finding indicates that GSK-3β inhibition may be a potential therapeutic target to control NAFLD.